

Week 7: Visualizing Information

October 7, 2026

WHY DESIGN MATTERS

“Having word processing software doesn’t make us great writers.”

— Stephen Few

ggplot (or an agent) will draw any chart you ask for. Whether it *works* depends on how human perception reads it. A chart can be **ugly** (poor aesthetics), **bad** (hard to perceive accurately), or **wrong** (misleading about the data). This week is about the rules behind good charts, so you can judge one, yours or an agent’s.

THE VISUAL-MEMORY SYSTEM

A (very) simplified model: what you see lands in **iconic memory** (fast, automatic), a little of it moves into **working memory** (slow, small, effortful), and only some of that reaches **long-term memory**. Effective charts do two things:

1. **Grab and direct attention** (iconic memory).
2. **Reduce processing demands** (working memory).

Pre-attentive attributes

Some features pop out before you consciously search: color, size, shape, position, orientation. Counting every “5” in a block of digits is slow; it’s instant once the 5s are colored. Use these attributes to point the eye at what matters.

- **Numerical (ratio) data:** position, length, size.
- **Categorical data:** hue beats shape. Picking out a red dot among blue dots is instant; among circles and triangles it’s slow.
- **Less is more.** A few categories stay in working memory; ten colors make the reader flip back and forth to the legend.

CLEVELAND’S PATTERN PERCEPTION

Cleveland & McGill (1985) name three operations a reader performs on a chart: **estimation**, **assembly**, and **detection**.

1. Estimation

Judging values: **discriminate** (is X equal to Y?), **rank** (is X greater than Y?), and **ratio** (is X double Y?). Encodings differ in how accurately they support each, from most to least accurate:

Encoding	Discriminate	Rank	Ratio
1. Position, common scale	Yes	Yes	Yes
2. Position, non-aligned	Partly	Partly	Yes
3. Length	Partly	Partly	Yes
4. Angle	Partly	Yes	Partly
5. Area	Partly	Partly	Partly
6. Color saturation	Partly	Partly	No
7. Color hue	Partly	Partly	No

- **Sorting helps every encoding.** Even a well-encoded dot plot is hard to read in alphabetical order.
- **Dots on a common scale don’t need a zero baseline.** Starting at zero lowers resolution and isn’t needed for ratioing. **Bars encode length, so bars must start at zero.**
- Area works okay for bubble charts (Gapminder), not for precise values.

2. Assembly: Gestalt psychology

Assembly is how we group graphical elements. *The whole has a reality entirely separate from the parts.*

Law	We tend to...	Chart use
Closure	Fill in missing information	Gaps read as shapes
Prägnanz	Prefer regular, simple, orderly	Sort the bars
Continuity	Group objects along a direction	Trend lines
Similarity	Group <i>similar-looking</i> objects	Color or shape groups
Proximity	Group <i>nearby</i> objects	Dodged bars, facets

3. Detection

Recognizing that a geometric object encodes a value. A **Norman door** gives the wrong signal and needs a sign to correct it. A chart that needs a paragraph of explanation is a Norman door.

Encode your most important comparison with the most accurate channel: position on a common scale, sorted. Hue, area, and angle are for secondary information. If the reader has to estimate a ratio from a pie slice or a color, the chart is working against them.

10 DATA VIZ BEST PRACTICES

#	Rule	Exception / note
1	Remove chart junk	“Erase non-data ink” (Tufte)
2	No 3D plots	Hardly ever
3	Don’t lie	Bars start at 0; no cherry-picking
4	No pie charts	OK: small data, simple fractions
5	Don’t stack bars	OK when the <i>total</i> matters most
6	Rotate & sort axes	Ordinal variables keep their order
7	Label geoms directly	Legend OK for repeated categories
8	No pattern fills	Use color or shade instead
9	No red with green	Color-blind readers
10	Tables for small data	A 2 × 2 is a table

Why these rules

- **3D:** we’re bad at judging depth, ink proportions stop matching true proportions, geoms occlude each other, and the third dimension distracts.
- **Lying:** truncated bar axes, cherry-picked time windows, and charts whose parts don’t add up (percentages over 100%).
- **Stacked bars:** only the bottom segment sits on a common scale. A parallel-coordinates (line) plot usually works better.
- **Sorting:** the default alphabetical order is almost always wrong. Order by value with `reorder()`.
- **Color blindness:** about 10% of males and 1% of females. Facets can avoid color altogether.

MAKING A (GOOD) GGPLOT

Seven steps, applied to the wildlife strikes data:

Step	What to do
1. Format data frame	<code>count()</code> , <code>filter()</code> , <code>mutate()</code>
2. Add geoms	<code>geom_col()</code> , <code>geom_point()</code> , ...
3. Flip coordinates?	Put long labels on the y axis
4. Reorder factors?	<code>reorder(operator, n)</code>
5. Adjust scales	Remove padding; set breaks
6. Adjust theme	A cowplot theme
7. Annotate	<code>labs()</code> for title and axes

```
wildlife_impacts %>%
  count(operator) %>%                                # 1
  mutate(operator = str_to_title(operator)) %>%
  ggplot() +
  geom_col(                                             # 2
    aes(x = n, y = reorder(operator, n)),              # 3, 4
    width = 0.7, alpha = 0.8
  ) +
  scale_x_continuous(                                  # 5
    expand = expansion(mult = c(0, 0.05))
  ) +
  theme_minimal_vgrid() +                              # 6
  labs(x = 'Count', y = NULL)                          # 7
```

- Flip by mapping the count to x and the category to y, rather than adding `coord_flip()`.
- `expand = expansion(mult = c(0, 0.05))` makes the bars touch the axis. Add `breaks = and limits =` to control tick marks.
- Four cowplot themes: `theme_half_open()`, `theme_minimal_grid()`, `theme_minimal_vgrid()`, `theme_minimal_hgrid()`. **Horizontal bars get only vertical grid lines** (`vgrid`); vertical bars get `hgrid`.

The practice file repeats this with `gapminder.csv`: life expectancy in 2007 for countries in the Americas, sorted, as horizontal bars.

CRITIQUING A CHART

The in-class exercise, and a habit for every chart an agent hands you:

1. Where does it fall on Cleveland's hierarchy? Which encoding carries the main comparison?
2. Which design rules does it break?
3. Suggest at least two improvements.

DIRECTING AN AGENT TO PLOT

An agent defaults to ggplot defaults: alphabetical order, legends, gray backgrounds. Ask for the design, then check it against the rules above:

- *"Make horizontal bars of strike counts by operator, sorted by count, with bars starting at the axis and only vertical grid lines."*
- *"Replace the legend with direct labels at the end of each line."*
- *"This is a stacked bar chart. Redo it as dodged bars or a line plot so each category sits on a common scale."*
- *"Check this palette for red-green color-blind readers."*

CHART SYMPTOMS & FIXES

Symptom	Fix
Bars in alphabetical order	<code>reorder(operator, n)</code>
Overlapping category labels	Map the category to y
Gap between bars and axis	Set <code>expand</code> in the scale
Gray background, heavy grid	A cowplot theme
Eyes bounce to the legend	Direct labels; drop the legend
Too many colors to track	Fewer groups, or facets
Bar axis starts above 0	Start at 0, or switch to dots
Parts hard to compare	Dodged bars, not stacked or pie

BEFORE NEXT CLASS

- **HW6: Factors, Amounts, & Proportions**, due **Oct 12**.
- **MP2: Explore & Interrogate**, due **Oct 25** (9% of grade).
- **Quiz 3** on Oct 14, covering Weeks 6–7.
- References: [data viz cheat sheet](#) and [data viz reference page](#).

You should be able to:

- Explain the two objectives of an effective chart.
- Rank encodings on Cleveland's hierarchy and say which support ratioing.
- Name the five Gestalt laws and spot them in a chart.
- Apply the 10 best practices and their exceptions.
- Build a sorted, cleanly themed bar chart in seven steps.
- Critique a chart and propose concrete improvements.